

M71 - Mean reading length vs switch-rate (correlation)

Clean questions: 172 Participants (avg across both Qs): 101

Level	Subset	Reading	Switch	n	Pear r	Pear p	Spear r	Spear p
per_question	all	mean_reading_length_s	switch_rate_typing	172	+0.143	0.0608	+0.257	0.000674
per_question	all	mean_reading_length_s	switch_rate_topic	172	-0.092	0.228	-0.114	0.138
per_question	all	mean_reading_length_s	switch_rate_time	172	+0.168	0.0277	+0.131	0.0871
per_question	all	median_reading_length_s	switch_rate_typing	172	+0.101	0.187	+0.286	0.000141
per_question	all	median_reading_length_s	switch_rate_topic	172	+0.021	0.786	+0.019	0.805
per_question	all	median_reading_length_s	switch_rate_time	172	+0.134	0.0804	+0.165	0.0308
per_participant	all	mean_reading_length_s	switch_rate_typing	101	+0.024	0.812	+0.177	0.0761
per_participant	all	mean_reading_length_s	switch_rate_topic	101	-0.151	0.131	-0.144	0.151
per_participant	all	mean_reading_length_s	switch_rate_time	101	+0.313	0.00143	+0.182	0.0684
per_participant	all	median_reading_length_s	switch_rate_typing	101	-0.037	0.716	+0.236	0.0176
per_participant	all	median_reading_length_s	switch_rate_topic	101	-0.030	0.765	+0.000	1
per_participant	all	median_reading_length_s	switch_rate_time	101	+0.269	0.00661	+0.208	0.0373
per_participant	diffuse	mean_reading_length_s	switch_rate_typing	52	+0.330	0.0167	+0.270	0.0532
per_participant	diffuse	mean_reading_length_s	switch_rate_topic	52	-0.127	0.371	-0.025	0.862
per_participant	diffuse	mean_reading_length_s	switch_rate_time	52	+0.112	0.428	+0.028	0.845
per_participant	clumpy	mean_reading_length_s	switch_rate_typing	49	-0.205	0.157	+0.078	0.595
per_participant	clumpy	mean_reading_length_s	switch_rate_topic	49	-0.173	0.234	-0.161	0.27
per_participant	clumpy	mean_reading_length_s	switch_rate_time	49	+0.434	0.00184	+0.283	0.0488

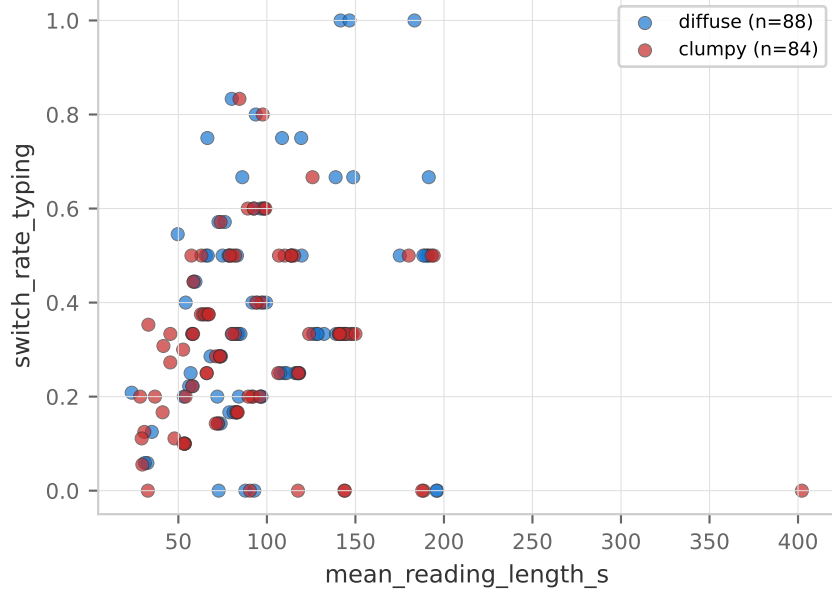
Interpretation:

$\text{switch_rate} = n_switches / (n_pages - 1).$

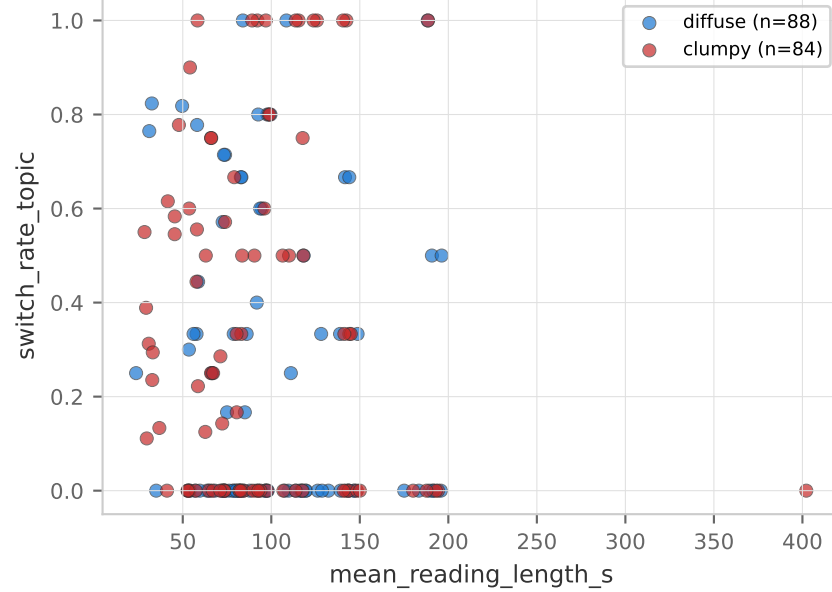
Strong negative correlation = longer dwell times go with fewer alternations,
which would suggest the two measures partly redundantly index "exploit" behavior.

M71 - Mean reading length vs switch rate (scatter)

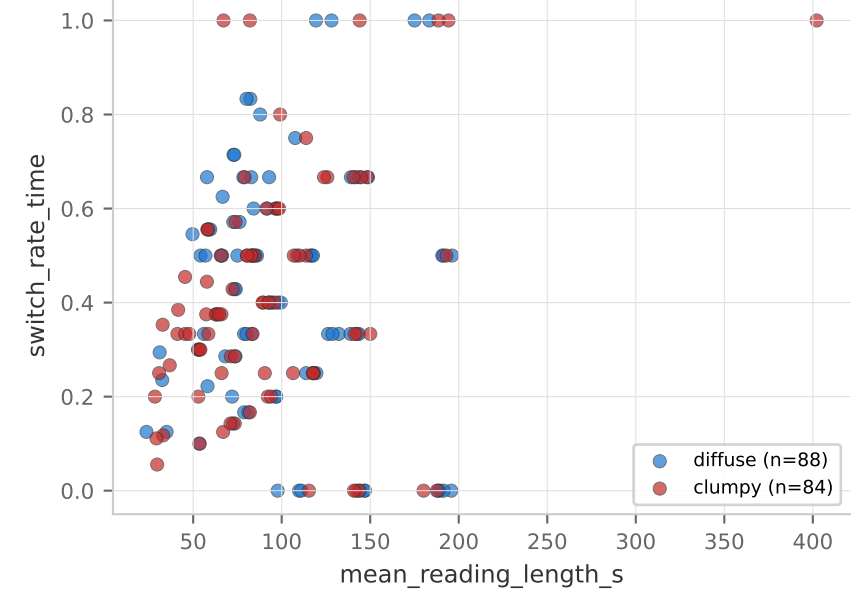
Per question: reading vs typing-switch
Pearson $r=+0.14$ $p=0.0608$ Spearman $r=+0.26$ $p=0.000674$



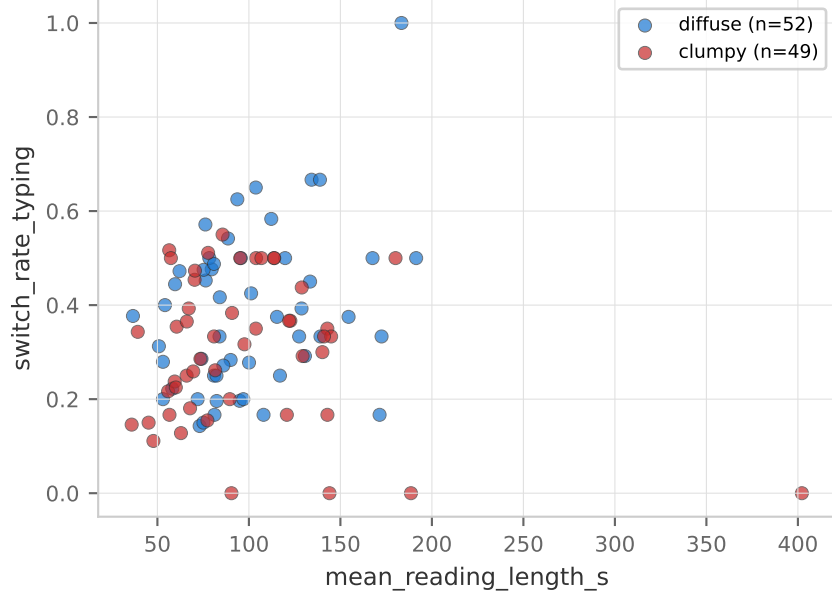
Per question: reading vs topic-switch
Pearson $r=-0.09$ $p=0.228$ Spearman $r=-0.11$ $p=0.138$



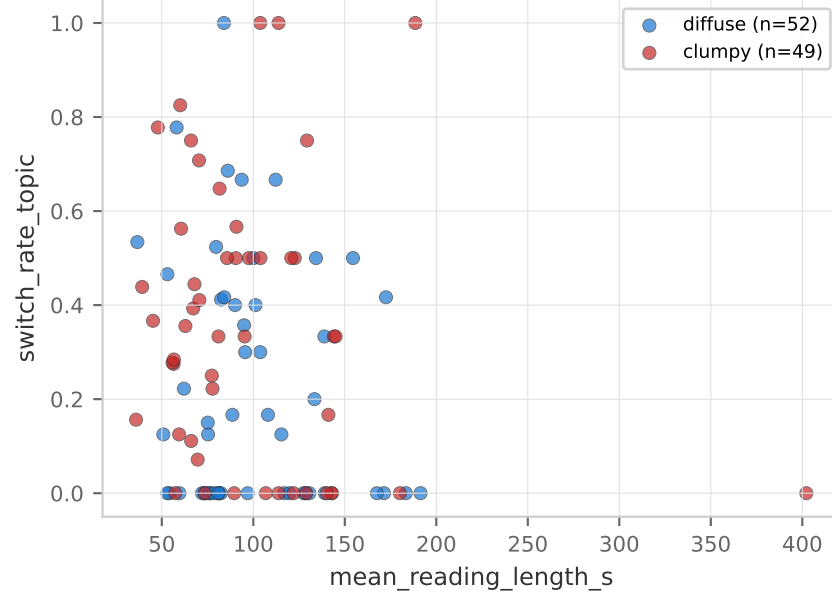
Per question: reading vs time-switch
Pearson $r=+0.17$ $p=0.0277$ Spearman $r=+0.13$ $p=0.087$



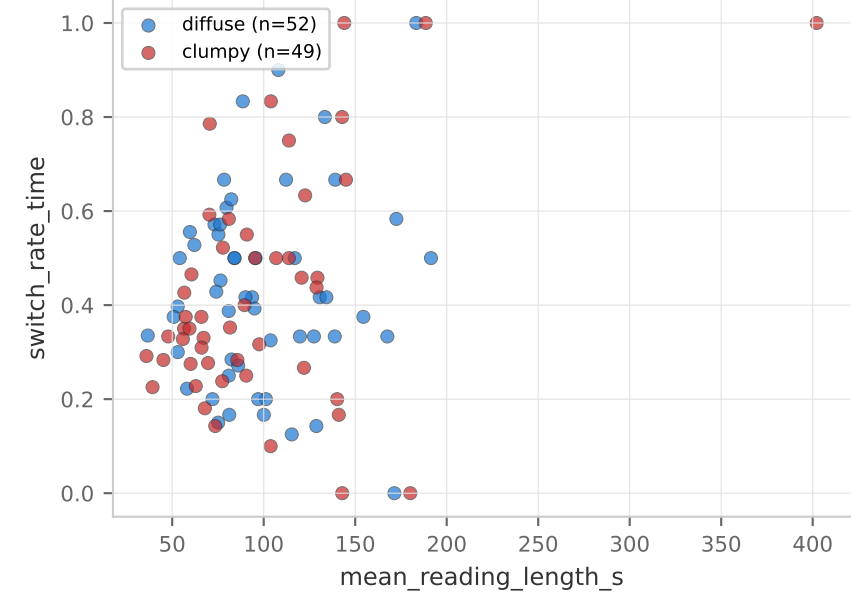
Per participant: reading vs typing-switch
Pearson $r=+0.02$ $p=0.812$ Spearman $r=+0.18$ $p=0.0761$



Per participant: reading vs topic-switch
Pearson $r=-0.15$ $p=0.131$ Spearman $r=-0.14$ $p=0.151$



Per participant: reading vs time-switch
Pearson $r=+0.31$ $p=0.00143$ Spearman $r=+0.18$ $p=0.068$



M71 - Bottom line

Sample: N=101 participants (172 clean questions). Per-participant level (averaged across both questions).

HEADLINE

Mean reading length and the TYPING-based switch rate are
INDEPENDENT at the participant level - they measure DIFFERENT things.

PER MEASURE (reading length vs ...)

typing (primary) $r=+0.02$ ($p=0.81$); $\rho=+0.18$ ($p=0.08$)

-> *Independent. Keep as separate construct.*

topic (LDA) $r=-0.15$ ($p=0.13$); $\rho=-0.14$ ($p=0.15$)

-> *Weak negative, not significant. Also independent.*

time (>60s) $r=+0.31$ ($p=0.00$); $\rho=+0.18$ ($p=0.07$)

-> *Positive, MECHANICAL (both derived from dwell time).*

BY CONDITION (typing-switch only)

diffuse $r=+0.33$ ($p=0.02$); $\rho=+0.27$ ($p=0.05$)

-> *n=52 - longer readers switch slightly more*

clumpy $r=-0.21$ ($p=0.16$); $\rho=+0.08$ ($p=0.59$)

-> *n=49 - no positive link*

TALKING POINTS

"Reading length and typing-switch are independent at the participant level - they are NOT measuring the same thing."

"The only sizeable correlation (with time-switch, $r=+.31$) is mechanical: both are computed from page dwell time."

"Topic-switch is also essentially independent of reading length."